

# Image *and* Video Compression *for* Multimedia Engineering

*Fundamentals, Algorithms, and Standards*

*Second Edition*

Yun Q. Shi  
Huifang Sun



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Yun Q. Shi

*New Jersey Institute of Technology  
Newark, New Jersey, USA*

Huifang Sun

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*To beloved Kong Wai Shih, Wen Su,*

*Yi Xi Li, Shu Jun Zheng, and*

*Xian Hong Li*

*and*

*To beloved Xuedong,*

*Min, Yin, Andrew, Rich, Haixin, and*

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## *Preface to the Second Edition*

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When looking at the preface of the first edition of this book published in 1999, it is observed that most of the presentation, analyses, and discussion made there are still valid. The trend of switching from analog to digital communications continues. Digital image and video, digital multimedia, Internet, and World Wide Web have been continuously and vigorously growing during the past eight years. Therefore, in this second edition of this book, we have retained most of the material of the first edition, but with some necessary updates and new additions. Two major and some minor changes made in this second edition are as follows.

First, the major parts of JPEG2000 have become standards after 1999. Hence, we have updated Chapter 8, which presents fundamental concepts and algorithms on JPEG2000. Second, a new chapter describing the recently developed video coding standard, MPEG-4 Part 10 Advanced Video Coding or H.264, has been added in the second edition as Chapter 20. For this purpose, Chapter 20 in the first edition covering the system part of MPEG, multiplexing/demultiplexing and synchronizing the coded audio, video, and other data has been changed to Chapter 21 in this new edition. Other minor changes have been made wherever necessary, including the addition of new color systems of digital video, profiles, and levels of video coding standards.

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## *Preface to the First Edition*

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It is well known that in the 1960s the advent of the semiconductor computer and the space program swiftly brought the field of digital image processing into public focus. Since then the field has experienced rapid growth and has entered every aspect of modern technology. Since the early 1980s, digital image sequence processing has been an attractive research area because an image sequence, as a collection of images, may provide more information than a single image frame. The increased computational complexity and memory space required for image sequence processing are becoming more attainable. This is due to more advanced, achievable computational capability, resulting from the continuing progress made in technologies, especially those associated with the VLSI industry and information processing.

In addition to image and image sequence processing in the digitized domain, facsimile transmission has switched from analog to digital since the 1970s. However, the concept of high definition television (HDTV) when proposed in the late 1970s and early 1980s continued to be analog. This has since changed. In the United States, the first digital system proposal for HDTV appeared in 1990. The Advanced Television Standards Committee (ATSC), formed by the television industry, recommended the digital HDTV system developed jointly by the seven Grand Alliance members as the standard, which was approved by the Federal Communication Commission (FCC) in 1997. Today's worldwide prevailing concept of HDTV is digital. The digital television (DTV) provides the signal that can be used in computers. Consequently, the marriage of TV and computers has begun. Direct broadcasting by satellite (DBS), digital video disks (DVD), video-on-demanding (VOD), video game and other digital video related media and services are now, or soon to be, available.

As in the case of image and video transmission and storage, audio transmission and storage through some media have changed from analog to digital. Examples include entertainment audio on compact disks (CD) and telephone transmission over long and medium distance. Digital TV signals discussed above provide another example since they include audio signals. Transmission and storage of audio signals through some other media are about to change to digital. Examples of this include telephone transmission through local area and cable TV.

Although most signals generated from various sensors are analog in nature, the switching from analog to digital is motivated by the superiority of digital signal processing and transmission over their analog counterparts. The principal advantage of being digital is the robustness against various noises. Clearly, this results from the fact that only binary digits exist in digital format and it is much easier to distinguish one state from the other than to handle analog signals.

Another advantage of being digital is ease of signal manipulation. In addition to the development of a variety of digital signal (including image, video and audio) processing techniques and specially designed software and hardware, that may be well-known, the following development is an example of this advantage. The digitized information format, i.e., the bitstream, often in the compressed version, is a revolutionary change in the video industry that enables many manipulations, which are either impossible or very complicated to execute in analog format. For instance, video, audio and other data can be first compressed



to separate bitstreams and then combined to a signal bitstream, thus providing a multimedia solution for many practical applications. Information from different sources and to different devices can be multiplexed and demultiplexed in terms of the bitstream. Bitstream conversion in terms of bit rate conversion, resolution conversion and syntax conversion becomes feasible. In digital video, content-based coding, retrieval and manipulation; and editing video in the compressed domain become feasible. All system-timing signals in the digital systems can be included in the bitstream instead of being transmitted separately as in traditional analog systems.

Being digital is well-suited to the recent development of modern telecommunication structures as exemplified by the Internet and World Wide Web (WWW). Therefore, we can see that digital computers, consumer electronics (including television and video games), and telecommunications networks are combined to produce an information revolution. Combining audio, video and other data, multimedia becomes an indispensable element of modern life. While the pace and the future of this revolution cannot be predicted, one thing is certain: this process is going to drastically change many aspects of our world in the next several decades.

One of the enabling technologies in the information revolution is digital data compression, because the digitization of analog signals causes data expansion. In other words, store and/or transmit digitized signals need more bandwidth and/or storage space than the original analog signals do.

The focus of this book is on image and video compression encountered in multimedia engineering. Fundamentals, algorithms and standards are three emphases of the book. It is intended to serve as a graduate level text. Its material is sufficient for a one-semester or one-quarter graduate course on digital image and video coding. For this purpose, at the end of each chapter, there is a section of exercises, containing problems and projects, for practice, and a section of references for further reading.

Based on this book, a short course, entitled "Image and Video Compression for Multimedia," was conducted at Nanyang Technological University, Singapore in March and April, 1999. The response to the short course was overwhelmingly positive.

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## *Content and Organization of the Book*

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This book consists of 21 chapters, grouped into four parts: (1) fundamentals, (2) still image compression, (3) motion estimation and compensation, and (4) video compression. The following paragraphs summarize the aim and content of each chapter, each part, and the relationship between some chapters and the four parts.

Part I includes Chapters 1–6, which provide readers with the fundamentals for understanding the remaining three parts of the book. In Chapter 1, the practical needs for image and video compression are demonstrated. The feasibility of image and video compression is analyzed. Specifically, both statistical and psychovisual redundancies are analyzed and the removal of these redundancies leads to image and video compression. In the course of the analysis, some fundamental characteristics of the human visual system are discussed. Visual quality measurement, another important concept in the compression, is addressed in both subjective and objective quality measures, and the new trend in combining the merits of the two measures is also discussed. Finally, some information theory results are presented as the concluding subject of the chapter.

Chapter 2 discusses quantization, a crucial step in lossy compression. It is known that quantization has a direct impact on both coding bit rate and quality of reconstructed frames. Both uniform and nonuniform quantizations are covered in this chapter. The issues of quantization distortion, optimum quantization, and adaptive quantization are also addressed. The last subject discussed in the chapter is pulse code modulation (PCM), which, as the earliest, best-established, and most frequently applied coding system, normally serves as a standard against which other coding techniques are compared.

Two efficient coding schemes, differential coding and transform coding (TC), are discussed in Chapters 3 and 4, respectively. Both techniques utilize the redundancies discussed in Chapter 1, thus achieving data compression. In Chapter 3, the formulation of general differential pulse code modulation (DPCM) systems is described first, followed by the discussion of optimum linear prediction and several implementation issues. Then, delta modulation (DM), as an important, simple, special case of DPCM is presented. Finally, application of differential coding technique to interframe coding and information preserving differential coding are covered.

Chapter 4 begins with the introduction of the Hotelling transform, the discrete version of the optimum Karhunen and Loeve transform. Through the statistical, geometrical, and basis vector (image) interpretations, this introduction provides a solid understanding of the transform coding technique. Several linear unitary transforms are then presented, followed by performance comparisons between these transforms in terms of energy compactness, mean square reconstruction error, and computational complexity. It is demonstrated that the discrete cosine transform (DCT) performs better than others in general. In the discussion of bit allocation, an efficient adaptive scheme using thresholding coding devised by Chen and Pratt in 1984 is featured, which established a basis for the international still image coding standard, JPEG. The comparison between DPCM and TC is also given. The combination of these two techniques (hybrid transform/waveform coding), and its application in image and video coding are also described.

The last two chapters in Part I cover several coding (e.g., code word assignment) techniques. In Chapter 5, two types of variable-length coding techniques, Huffman coding and arithmetic coding, are discussed. First, an introduction to basic coding theory is presented, which can be viewed as a continuation of the information theory results presented in Chapter 1. Then, the Huffman code, as an optimum and instantaneous code, and a modified version are covered. Huffman coding is a systematic procedure for encoding a source alphabet with each source symbol having an occurrence probability. As a block code (a fixed code word having an integer number of bits is assigned to a source symbol), it is optimum in the sense that it produces minimum coding redundancy. Some limitations of Huffman coding are analyzed. As a stream-based coding technique, arithmetic coding is distinct from and is gaining more popularity than Huffman coding. It maps a string of source symbols into a string of code symbols. Free of the integer-bits-per-source-symbol restriction, arithmetic coding is more efficient. The principle of arithmetic coding and some of its implementation issues are addressed.

While two types of variable-length coding techniques, introduced in Chapter 5, can be classified as fixed-length-to-variable-length coding techniques, run-length coding (RLC) and dictionary coding (as discussed in Chapter 6) can be classified as variable-length-to-fixed-length coding techniques. The discrete Markov source model (another portion of the information theory results), which can be used to characterize 1-D RLC, is introduced at the beginning of Chapter 6. Both 1-D RLC and 2-D RLC are then introduced. The comparison between 1-D and 2-D RLC is made in terms of coding efficiency and transmission error effect. The digital facsimile coding standards based on 1-D and 2-D RLC are introduced. Another focus of Chapter 6 is on dictionary coding. Two groups of adaptive dictionary coding techniques, the LZ77 and LZ78 algorithms, are presented. Their applications are discussed. At the end of the chapter, a discussion of international standards for lossless still image compression is given. For both lossless bilevel and multi-level still image compression, the respective standard algorithms and their performance comparisons are provided.

Part II of the book includes Chapters 7 through 9, which are devoted to still image compression. In Chapter 7, the international still image coding standard JPEG is introduced. Two classes of encoding, i.e., lossy and lossless, and four modes of operation, i.e., sequential DCT-based mode, progressive DCT-based mode, lossless mode, and hierarchical mode, are covered. The discussion in Part I is very useful in understanding what is introduced here for JPEG.

Because of the higher coding efficiency and superior spatial and quality scalability features over the DCT coding technique, discrete wavelet transform (DWT) coding has been adopted by JPEG2000 still image coding standards as the core technology. Chapter 8 begins with an introduction to wavelet transform (WT), which includes a comparison between WT and the short-time Fourier transform (STFT), and presents WT as a unification of several existing techniques, known as filter bank analysis, pyramid coding, and sub-band coding. Then the DWT for still image coding is discussed. In particular, the embedded zerotree wavelet (EZW) technique and set partitioning in hierarchical trees (SPIHT) are discussed. The updated JPEG2000 standard activity is also presented here.

Chapter 9 presents three nonstandard still image coding techniques: vector quantization (VQ), fractal coding, and model-based image coding. All three techniques have several important features such as very high compression ratio for certain kinds of images, and very simple decoding procedures. Owing to some limitations, however, they have not been adopted by the still image coding standards. On the other hand, the facial model and face animation techniques have been adopted by the MPEG-4 video standard.

Part III of this book, consisting of Chapters 10 through 14, addresses motion estimation and motion compensation, which are key issues in modern video compression. Part III is a

prerequisite to Part IV, which discusses various video coding standards. The first chapter in Part III, Chapter 10, introduces motion analysis and compensation in general. The chapter begins with the concept of imaging space, which characterizes all images and all image sequences in temporal and spatial domains. Both temporal and spatial image sequences are special proper subsets of the imaging space. A single image becomes merely a specific cross section of the imaging space. Two techniques in video compression utilizing interframe correlation, both developed in the late 1960s and early 1970s, are presented here. Frame replenishment is relatively simpler in modeling and implementation. However, motion compensated coding achieves higher coding efficiency and better quality in reconstructed frames with a 2-D displacement model. Motion analysis is then viewed from a signal processing perspective. Three techniques in motion analysis are briefly discussed. They are block matching, pel recursion, and optical flow, which are presented in detail in Chapters 11 through 13, respectively. Finally, other applications of motion compensation to image sequence processing are discussed.

Chapter 11 addresses the block matching technique, which is presently the most frequently used motion estimation technique. The chapter first presents the original block matching technique proposed by Jain and Jain. Several different matching criteria and search strategies are then discussed. A thresholding multiresolution block matching algorithm is described in some detail so as to provide an insight into the technique. Then, the limitations of block matching techniques are analyzed, from which several new improvements are presented. They include hierarchical block matching, multigrid block matching, predictive motion field segmentation, and overlapped block matching. All of these techniques modify the nonoverlapped, equally spaced, fix-sized, small rectangular block model proposed by Jain and Jain in some way so that the motion estimation is more accurate and has fewer block artifacts and overhead side information.

The pel recursive technique is discussed in Chapter 12. First, determination of 2-D displacement vectors is converted via the use of the displaced frame difference (DFD) concept to a minimization problem. Second, descent methods in optimization theory are discussed. In particular, the steepest descent method and Newton-Raphson method are addressed in terms of algorithm, convergence, and implementation issues such as selection of step-size and initial value. Third, the first pel recursive techniques proposed by Netravali and Robbins are presented. Finally, several improvement algorithms are described.

Optical flow, the third technique in motion estimation for video coding, is covered in Chapter 13. First, some fundamental issues in motion estimation are addressed. They include the difference and relationships between 2-D motion and optical flow, the aperture problem, and the ill-posed nature of motion estimation. The gradient-based and correlation-based approaches to optical flow determination are then discussed in detail. For the former, the Horn and Schunck algorithm is illustrated as a representative technique and some other algorithms are briefly introduced. For the latter, the Singh method is introduced as a representative technique. In particular, the concepts of conservation information and neighborhood information are emphasized. A correlation-feedback algorithm is presented in detail to provide an insight into the correlation technique. Finally, multiple attributes for conservation information are discussed.

Chapter 14, the last chapter in Part III, provides a further discussion and summary of 2-D motion estimation. First, a few features common to all three major techniques discussed in Chapters 11 through 13 are addressed. They are the aperture and ill-posed inverse problems, conservation and neighborhood information, occlusion and disocclusion, and rigid and nonrigid motion. Second, a variety of different classifications of motion estimation techniques are presented. Frequency domain methods are discussed as well. Third, performance comparison between three major techniques in motion estimation is made. Finally, the new trends in motion estimation are presented.

Part IV, containing Chapters 15 through 21, covers various video coding standards. Chapter 15 presents fundamentals of video coding. First, digital video representation is discussed. Second, the rate distortion function of the video signal is covered, the fourth portion of the information theory results presented in this book. Third, various digital video formats are discussed. Finally, the current digital image/video coding standards are summarized. The full names and abbreviations of some organizations, the completion time, and the major features of various image/video coding standards are listed in two tables.

Chapter 16 is devoted to video coding standards MPEG-1/2, which are the most widely used video coding standards at present. The basic technique of MPEG-1/2 is a full-motion compensated DCT and DPCM hybrid coding algorithm. The features of MPEG-1 (including layered data structure) and the MPEG-2 enhancements (including field/frame modes for supporting the interlaced video input and scalability extension) are described. Issues of rate control, optimum mode decision, and multiplexing are discussed.

Chapter 17 presents several application examples of MPEG-1/2 video standards. They are the ATSC DTV standard, approved by the Federal Communications Commission (FCC) in the United States, transcoding, down-conversion decoder, and error concealment. Discussion of these applications can enhance understanding and mastering of MPEG-1/2 standards. Some research work reported may be found helpful for graduate students to broaden their knowledge of digital video processing, an active research field.

Chapter 18 presents the MPEG-4 video standard. The predominant feature of MPEG-4, content-based manipulation, is emphasized. The underlying concept of audio/visual objects (AVOs) is introduced. The important functionalities of MPEG-4, i.e., content-based interactivity (including bitstream editing, synthetic and natural hybrid coding (SNHC)), content-based coding efficiency, and universal access (including content-based scalability), are discussed. Since neither MPEG-1 nor MPEG-2 includes synthetic video and content-based coding, the most important application of MPEG-4 is in a multimedia environment.

Chapter 19 introduces ITU-T video coding standards H.261 and H.263, which are utilized mainly for videophony and videoconferencing. The basic technical detail of H.261, the earliest video coding standard, is presented. The technical improvements with which H.263 achieves high coding efficiency are discussed. Features of H.263+, H.263++, and H.26L are also presented.

Chapter 20 introduces the recently developed video coding standard, MPEG-4 Part 10 Advanced Video Coding [H.264], which is jointly developed by joint video team (JVC) of MPEG and ITU-T VCEG, and it is simply called as H.264/AVC. The H.264/AVC is an efficient and state-of-the-art video compression standard, whose coding efficiency is about two times better than that of MPEG-2 at the expense of increased complexity. The H.264/AVC has been planned for many applications including HD-DVD, DTV for satellite and wireless networks, IPTV, and many others.

Chapter 21 covers the systems part of MPEG, multiplexing/demultiplexing and synchronizing the coded audio, video and other data. Specifically, MPEG-2 systems and MPEG-4 systems are introduced. In MPEG-2 systems, two forms i.e., program stream and transport stream, are described. In MPEG-4 systems, some multimedia application related issues are discussed.

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and digital communication. He has coauthored two books and has published more than 150 journal and conference papers. He holds 48 U.S. patents. He received the technical achievement award in 1994 at the Sarnoff laboratory. He received the 1992 best paper award of *IEEE Transactions on Consumer Electronics*, the 1996 best paper award of ICCE, and the 2003 best paper award of *IEEE Transactions on Circuits and Systems for Video Technology*. Dr. Sun is now an associate editor for *IEEE Transactions on Circuits and Systems for Video Technology* and was the chair of the Visual Processing Technical Committee of IEEE Circuits and System Society. He is an IEEE Fellow.

# **Part I**

## **Fundamentals**



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# 1

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## *Introduction*

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Image and video data compression refers to a process in which the amount of data used to represent image and video is reduced to meet a bit rate requirement (below or at most equal to the maximum available bit rate), while the quality of the reconstructed image or video satisfies a requirement for a certain application and the complexity of computation involved is affordable for the application. In this book, the terms image and video data compression, image and video compression, and image and video coding are synonymous. Figure 1.1 shows the functionality of image and video data compression in visual transmission and storage. Image and video data compression has been found to be necessary in these important applications, because the huge amount of data involved in these and other applications usually well-exceeds the capability of today's hardware despite rapid advancements in semiconductor, computer, and other industries.

It is noted that both information and data are closely related yet different concepts. Data represents information and the quantity of data can be measured. In the context of digital image and video, data is usually measured in the number of binary units (bits). Information is defined as knowledge, facts, and news according to the Cambridge International Dictionary of English. That is, while data is the representation of knowledge, facts, and news, information is the knowledge, facts, and news. Information, however, may also be quantitatively measured.

Bit rate (also known as coding rate), as an important parameter in image and video compression, is often expressed in a unit of bits per second (bits/s, or bps), which is suitable in visual communication. In fact, an example in Section 1.1 concerning videophony (a case of visual transmission) uses bit rate in terms of bits per second. In the application of image storage, bit rate is usually expressed in a unit of bits per pixel (bpp). The term pixel is an abbreviation for picture element and is sometimes referred to as pel. In information source coding, bit rate is sometimes expressed in a unit of bits per symbol. In Section 1.4.2, when discussing noiseless source coding theorem, we consider bit rate as the average length of code words in the unit of bits per symbol.

The required quality of the reconstructed image and video is application dependent. In medical diagnosis and some scientific measurements, we may need the reconstructed image and video to mirror the original image and video. In other words, only reversible, information-preserving schemes are allowed. This type of compression is referred to as lossless compression. In applications, such as motion picture and television (TV), a certain amount of information loss is allowed. This type of compression is called lossy compression.

From its definition, one can see that image and video data compression involves several fundamental concepts including information, data, visual quality of image and video, and computational complexity. This chapter is concerned with several fundamental concepts in image and video compression. First, the necessity as well as the feasibility of image and video data compression are discussed. The discussion includes the utilization of several types of redundancy inherent in image and video data, and the visual perception of the

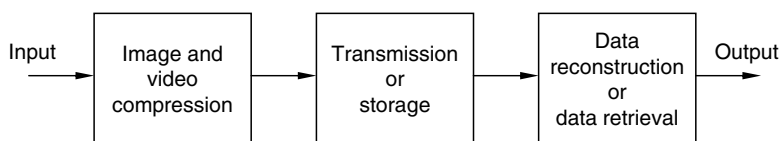
**FIGURE 1.1**

Image and video compression for visual transmission and storage.

human visual system (HVS). As the quality of the reconstructed image and video is one of our main concerns, the subjective and objective measures of visual quality are addressed. Then we present some fundamental information theory results, considering they play a key role in image and video compression.

## 1.1 Practical Needs for Image and Video Compression

Needless to say, visual information is of vital importance for human beings to perceive, recognize, and understand the surrounding world. With the tremendous progress that has been made in advanced technologies, particularly in very large-scale integrated (VLSI) circuits, increasingly powerful computers and computations, it is becoming more possible than ever for video to be widely utilized in our daily life. Examples include videophony, videoconferencing, high definition TV (HDTV), and digital video disk (also known as digital versatile disk [DVD]), to name a few.

Video as a sequence of video frames, however, involves a huge amount of data. Let us take a look at an illustrative example. Assume the present switch telephone network (PSTN) modem can operate at a maximum bit rate of 56,600 bits/s. Assume each video frame has a resolution of  $288 \times 352$  (288 lines and 352 pixels/line), which is comparable with that of a normal TV picture and is referred to as common intermediate format (CIF). Each of the three primary colors RGB (red, green, blue) is represented for one pixel with 8 bits, as usual, and the frame rate in transmission is 30 frames/s to provide a continuous motion video. The required bit rate, then, is  $288 \times 352 \times 8 \times 3 \times 30 = 72,990,720$  bits/s. Therefore, the ratio between the required bit rate and the largest possible bit rate is about 1289. This implies that we have to compress the video data by at least 1289 times in order to accomplish the transmission described in this example. Note that an audio signal has not been accounted for yet in this illustration.

With increasingly demanding video services, such as three-dimensional (3-D) movies and games, and high video quality, such as HDTV, advanced image, and video data compression is necessary. It becomes an enabling technology to bridge the gap between the required huge amount of video data and the limited hardware capability.

## 1.2 Feasibility of Image and Video Compression

In this section, we shall see that image and video compression is not only a necessity for rapid growth of digital visual communications, but is also feasible. Its feasibility rests with two types of redundancies, i.e., statistical redundancy and psychovisual redundancy. By eliminating these redundancies, we can achieve image and video compression.

### 1.2.1 Statistical Redundancy

Statistical redundancy can be classified into two types: interpixel redundancy and coding redundancy. By interpixel redundancy we mean that pixels of an image frame, and pixels of a group of successive image or video frames, are not statistically independent. On the contrary, they are correlated to various degrees. (Difference and relationship between image and video sequences are discussed in Chapter 10, when we begin to discuss video compression). This type of interpixel correlation is referred to as interpixel redundancy. Interpixel redundancy can further be divided into two categories: spatial redundancy and temporal redundancy. By coding redundancy, we mean that the statistical redundancy is associated with coding techniques.

#### 1.2.1.1 Spatial Redundancy

Spatial redundancy represents the statistical correlation between pixels within an image frame. Hence it is also called intraframe redundancy.

It is well known that for most properly sampled TV signals the normalized autocorrelation coefficients along a row (or a column) with a one-pixel shift is very close to the maximum value 1. That is, the intensity values of pixels along a row (or a column) have a very high autocorrelation (close to the maximum autocorrelation) with those of pixels along the same row (or the same column) but shifted by a pixel. This does not come as a surprise because most of the intensity values change continuously from pixel to pixel within an image frame except for the edge regions (Figure 1.2). Figure 1.2a is a pretty normal picture: a boy and a girl in a park, and is of a resolution of  $883 \times 710$ . The intensity profiles along the 318th row and the 262th column are depicted in Figure 1.2b and c, respectively. For easy reference, the positions of the 318th row and 262th column in the picture are shown in Figure 1.2d. That is, the vertical axis represents intensity values, while the horizontal axis indicates the pixel position within the row or the column. These two curves indicate that often intensity values change gradually from one pixel to the other along a row and along a column.

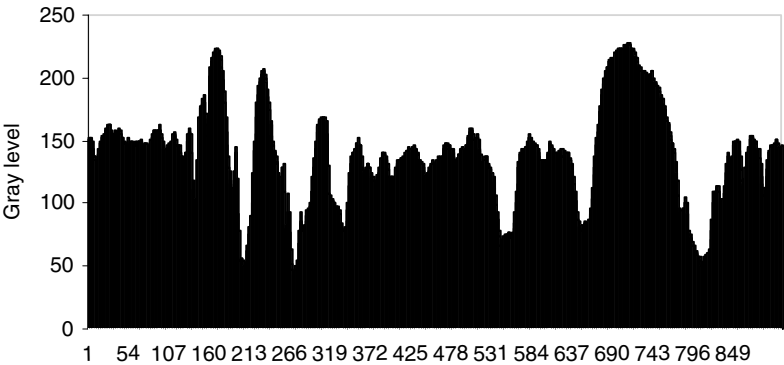
The study of the statistical properties of video signals can be traced back to the 1950s. Knowing that we must study and understand redundancy to remove it, Kretzmer designed some experimental devices such as a picture autocorrelator and a probabiloscope to measure several statistical quantities of TV signals and published his outstanding work in [kretzmer 1952]. He found that the autocorrelation in both horizontal and vertical directions exhibits similar behaviors as shown in Figure 1.3. Autocorrelation functions of several pictures with different complexity were measured. It was found that from picture to picture, the shape of autocorrelation curves ranges from remarkably linear to somewhat like exponential. The central symmetry with respect to the vertical axis and the bell-shaped distribution, however, remain generally the same. When the pixel shifting becomes small, it was found that the autocorrelation is high. This local autocorrelation can be as high as 0.97 to 0.99 for one- or two-pixel shifting. For very detailed pictures, it can range from 0.43 to 0.75. It was also found that autocorrelation generally has no preferred direction.

The Fourier transform of autocorrelation, power spectrum, is known as another important function in studying statistical behavior. Figure 1.4 shows a typical power spectrum of a TV signal [fink 1957; connor 1972]. It is reported that the spectrum is quite flat until 30 kHz for a broadcast TV signal. Beyond this line frequency the spectrum starts to drop at a rate of around 6 dB per octave. This reveals the heavy concentration of video signals in low frequencies, considering a nominal bandwidth of 5 MHz.

Spatial redundancy implies that the intensity value of a pixel can be guessed from that of its neighboring pixels. In other words, it is not necessary to represent each pixel in an image frame independently. Instead, one can predict a pixel from its neighbors. Predictive coding, also known as differential coding, is based on this observation and is discussed in Chapter 3. The direct consequence of recognition of spatial redundancy is that by removing a large amount of the redundancy (or utilizing the high correlation) within an image frame, we may save a lot of data in representing the frame, thus achieving data compression.

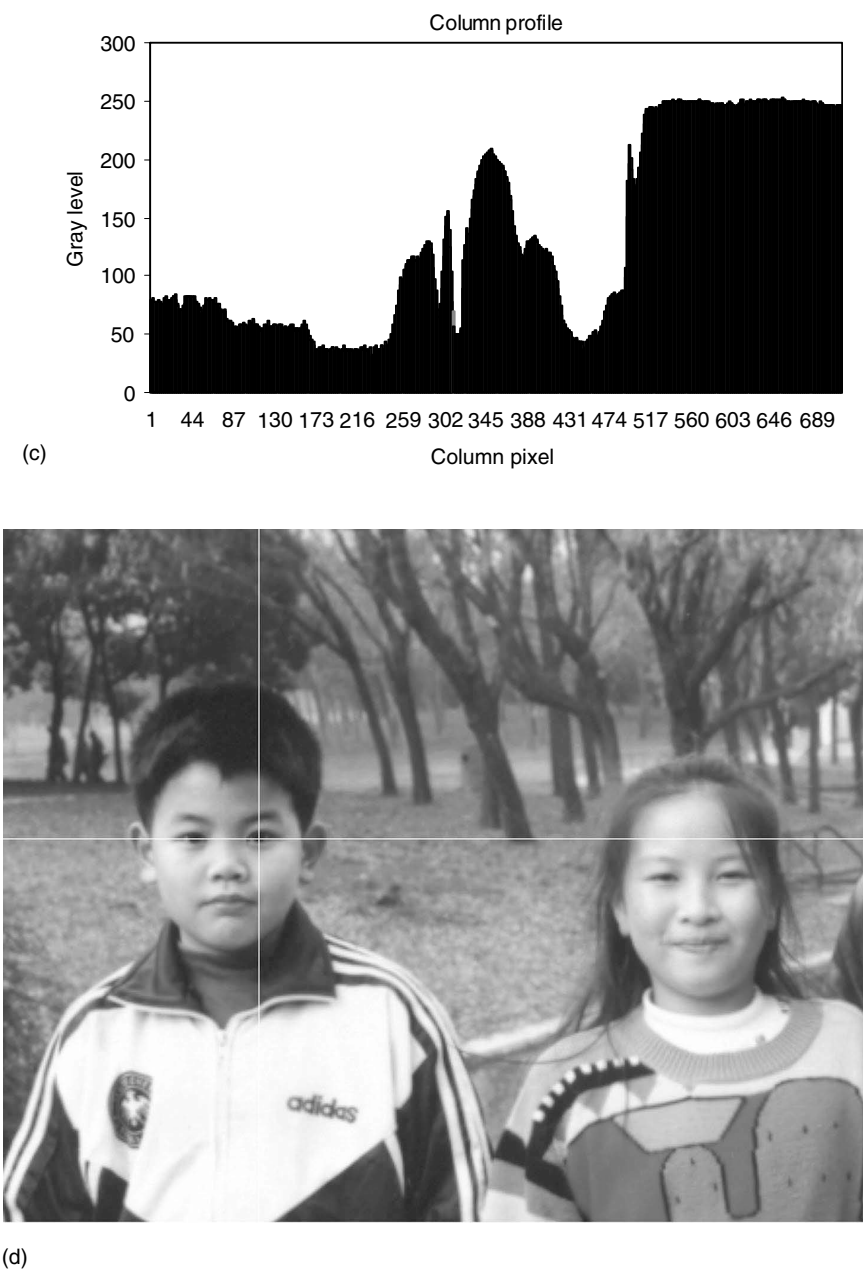


(a) Row profile



(b)

**FIGURE 1.2** (See color insert following page 288.)  
(a) A picture of boy and girl. (b) Intensity profile along 318th row.



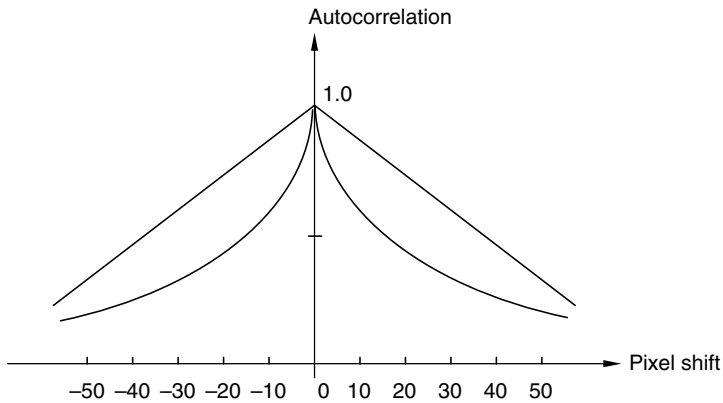
**FIGURE 1.2 (continued)**  
(c) Intensity profile along 262th column. (d) Positions of 318th row and 262th column.

**1.2.1.2 Temporal Redundancy**

Temporal redundancy is concerned with the statistical correlation between pixels from successive frames in a temporal image or video sequence. Therefore, it is also called interframe redundancy.

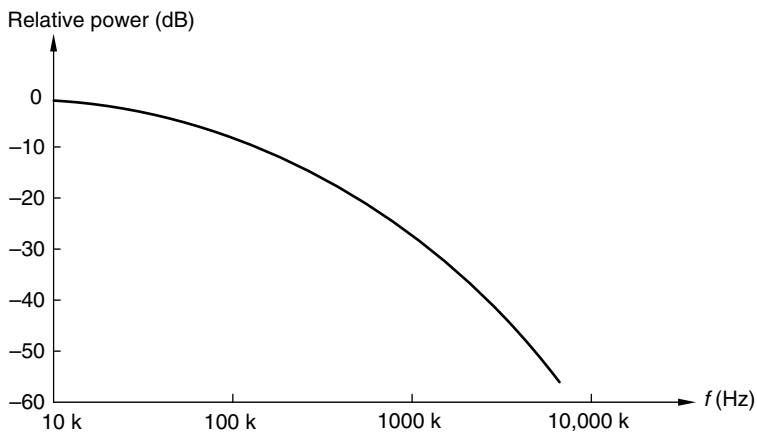
Consider a temporal image sequence. That is, a camera is fixed in the 3-D world and it takes pictures of the scene one by one as time goes by. As long as the time interval between two consecutive pictures is short enough, i.e., the pictures are taken densely enough, we can



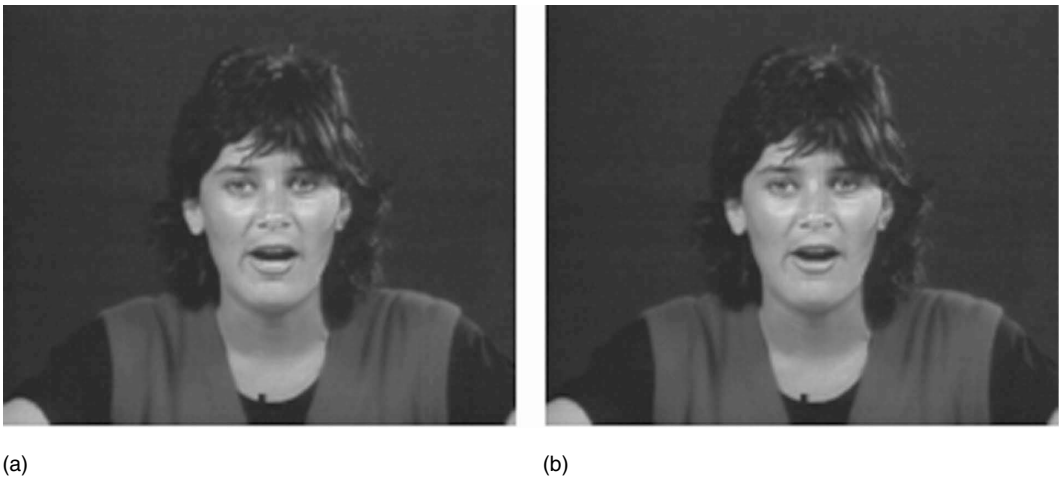


**FIGURE 1.3**  
Autocorrelation in horizontal directions for some testing pictures. (From Kretzmer, E.R., *Bell Syst. Tech. J.*, 31, 751, 1952. With permission.)

imagine that the similarity between two neighboring frames is strong. Figure 1.5a and b shows the 21st and 22nd frames of the “Miss America” sequence, respectively. The frames have a resolution of  $176 \times 144$ . Among the total 25,344 pixels, only 3.4% change their gray value more than 1% of the maximum gray value (255 in this case) from the 21st frame to the 22nd frame. This confirms an observation made in [mounts 1969]: For a videophone-like signal with moderate motion in the scene, on average, less than 10% of pixels change their gray values between two consecutive frames by an amount of 1% of the peak signal. The high interframe correlation was reported in [kretzmer 1952]. There, the autocorrelation between two adjacent frames was measured for two typical motion picture films. The measured autocorrelations were 0.80 and 0.86. The concept of frame difference coding of television signals was also reported in [seyler 1962], and the probability density functions of television frame differences was analyzed in [seyler 1965]. In summary, pixels within successive frames usually bear a strong similarity or correlation. As a result, we may predict a frame from its neighboring frames along the temporal dimension. This is referred to as interframe predictive coding and is discussed in Chapter 3. A more precise, hence, more efficient interframe predictive coding



**FIGURE 1.4**  
A typical power spectrum of a TV broadcast signal. (From Fink, D.G., *Television Engineering Handbook*, New York, 1957, Sect. 10.7. With permission.)



**FIGURE 1.5**  
(a) 21st frame and (b) 22nd frame of the Miss America sequence.

scheme, which has been in development since the 1980s, uses motion analysis. That is, it considers that the changes from one frame to the next are mainly due to the motion of some objects in the frame. Taking this motion information into consideration, we refer to the method as motion compensated (MC) predictive coding. Both interframe correlation and MC predictive coding are discussed in detail in Chapter 10.

Removing a large amount of temporal redundancy leads to a great deal of data compression. At present, all the international video coding standards have adopted MC predictive coding, which has been a vital factor to the increased use of digital video in digital media.

**1.2.1.3 Coding Redundancy**

As we discussed, interpixel redundancy is concerned with the correlation between pixels. That is, some information associated with pixels is redundant. The psychovisual redundancy (Section 1.2.2) is related to the information that is psychovisually redundant, i.e., to which the HVS is not sensitive. It is hence clear that both interpixel and psychovisual redundancies are somehow associated with some information contained in image and video. Eliminating these redundancies, or utilizing these correlations, by using fewer bits to represent the information results in image and video data compression. In this sense, the coding redundancy is different. It has nothing to do with information redundancy but with the representation of information, i.e., coding itself. To see this, let us take a look at the following example.

One illustrative example is provided in Table 1.1. The first column lists five distinct symbols that need to be encoded. The second column contains occurrence probabilities of

**TABLE 1.1**  
An Illustrative Example

| Symbol | Occurrence Probability | Code 1 | Code 2 |
|--------|------------------------|--------|--------|
| $a_1$  | 0.1                    | 000    | 0000   |
| $a_2$  | 0.2                    | 001    | 01     |
| $a_3$  | 0.5                    | 010    | 1      |
| $a_4$  | 0.05                   | 011    | 0001   |
| $a_5$  | 0.15                   | 100    | 001    |

these five symbols. The third column lists code 1, a set of code words obtained by using uniform-length code word assignment. (This code is known as the natural binary code.) The fourth column lists code 2, in which each code word has a variable length. Therefore, code 2 is called the variable-length code. It is noted that the symbol with a higher occurrence probability is encoded with a shorter length. Let us examine the efficiency of the two different codes. That is, we will examine which one provides a shorter average length of code words. It is obvious that the average length of code words in code 1,  $L_{\text{avg},1}$ , is 3 bits. The average length of code words in code 2,  $L_{\text{avg},2}$ , can be calculated as follows:

$$L_{\text{avg},2} = 4 \times 0.1 + 2 \times 0.2 + 1 \times 0.5 + 4 \times 0.05 + 3 \times 0.15 = 1.95 \text{ bits/symbol.} \quad (1.1)$$

Therefore, it is concluded that code 2 with variable-length coding is more efficient than code 1 with natural binary coding.

From this example, we can see that for the same set of symbols, different codes may perform differently. Some may be more efficient than others. For the same amount of information, code 1 contains some redundancy. That is, some data in code 1 is not necessary and can be removed without any effect. Huffman coding and arithmetic coding, two variable-length coding techniques, are discussed in Chapter 5.

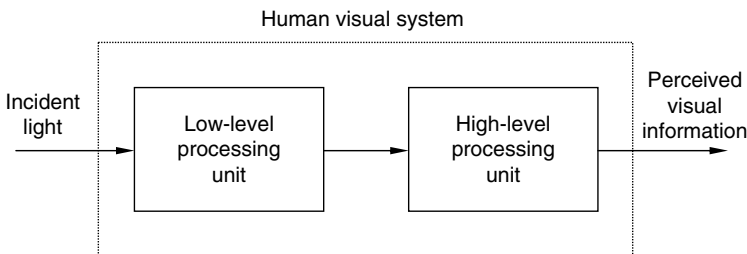
From the study of coding redundancy, it is clear that we should search for more efficient coding techniques to compress image and video data.

### 1.2.2 Psychovisual Redundancy

While interpixel redundancy inherently rests in image and video data, psychovisual redundancy originates from the characteristics of the HVS.

It is known that the HVS perceives the outside world in a rather complicated way. Its response to visual stimuli is a nonlinear function of the strength of some physical attributes of the stimuli such as intensity and color. HVS perception is different from camera sensing. In the VHS, visual information is not perceived equally; some information may be more important than other information. This implies that if we apply less data to represent less important visual information, perception will not be affected. In this sense, we see that some visual information is psychovisually redundant. Eliminating this type of psychovisual redundancy leads to data compression.

To understand this type of redundancy, let us study some properties of the HVS. We may model the human vision system as a cascade of two units [lim 1990], as depicted in Figure 1.6. The first one is a low-level processing unit that converts incident light into a neural signal. The second one is a high-level processing unit that extracts information from the neural signal. Although much research was carried out to investigate low-level processing, high-level processing remains wide open. The low-level processing unit is



**FIGURE 1.6**

A two-unit cascade model of the human visual system (HVS).

known as a nonlinear system (approximately logarithmic, as shown below). As a great body of literature exists, we limit our discussion only to video compression-related results. That is, several aspects of the HVS, which are closely related to image and video compression, are discussed in this section. They are luminance masking, texture masking, frequency masking, temporal masking, and color maskings. Their relevance in image and video compression is addressed. Finally, a summary is provided, in which it is pointed out that all of these features can be unified as one: differential sensitivity. This seems to be the most important feature of the human visual perception.

### 1.2.2.1 Luminance Masking

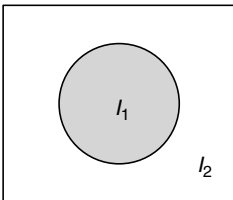
Luminance masking concerns the brightness perception of the HVS, which is the most fundamental aspect among the five to be discussed here. Luminance masking is also referred to as luminance dependence [connor 1972] and contrast masking [legge 1980; watson 1987]. As pointed in [legge 1980], the term masking usually refers to a destructive interaction or interference among stimuli that are closely coupled in time or space. This may result in a failure in detection, or errors in recognition. Here, we are mainly concerned with the detectability of one stimulus when another stimulus is present simultaneously. The effect of one stimulus on the detectability of another, however, does not have to decrease detectability. Indeed, there are some cases in which a low-contrast masker increases the detectability of a signal. This is sometimes referred to as facilitation, but in this discussion we only use the term masking.

Consider the monochrome image shown in Figure 1.7. There, a uniform disk-shaped object with a gray level (intensity value)  $I_1$  is imposed on a uniform background with a gray level  $I_2$ . Now the question is: Under what circumstances can the disk-shaped object be discriminated from the background by the HVS? That is, we want to study the effect of one stimulus (the background in this example, the masker) on the detectability of another stimulus (in this example, the disk). Two extreme cases are obvious. That is, if the difference between the two gray levels is quite large, the HVS has no problem with discrimination, or in other words the HVS notices the object from the background. If, on the other hand, the two gray levels are the same, the HVS cannot identify the existence of the object. What we are concerned with here is the critical threshold in the gray level difference for discrimination to take place.

If we define the threshold  $\Delta I$  as such a gray level difference  $\Delta I = I_1 - I_2$  that the object can be noticed by the HVS with a 50% chance, then we have the following relation, known as contrast sensitivity function, according to Weber's law.

$$\frac{\Delta I}{I} \approx \text{constant}, \quad (1.2)$$

where the constant is approximately 0.02. Weber's law states that for a relatively very wide range of  $I$ , the threshold for discrimination,  $\Delta I$ , is directly proportional to the intensity  $I$ . The implication of this result is that when the background is bright, a larger difference in gray levels is needed for the HVS to discriminate the object from the background.



**FIGURE 1.7**

A uniform object with gray level  $I_1$  imposed on a uniform background with gray level  $I_2$ .

On the other hand, the intensity difference required could be smaller if the background is relatively dark. It is noted that Equation 1.1 implies a logarithmic response of the HVS, and Weber's law holds for all other human senses as well.

Further research has indicated that the luminance threshold  $\Delta I$  increases more slowly than predicted by Weber's law. Some more accurate contrast sensitivity functions have been presented in the literature. In [legge 1980], it was reported that an exponential function replaces the linear relation in Weber's law. The following exponential expression is reported in [watson 1987].

$$\Delta I = I_0 \cdot \max \left\{ 1, \left( \frac{I}{I_0} \right)^\alpha \right\}, \quad (1.3)$$

where  $I_0$  is the luminance detection threshold when the gray level of the background is equal to zero, i.e.,  $I = 0$ , and  $\alpha$  is a constant, approximately equal to 0.7.

Figure 1.8 shows a picture uniformly corrupted by additive white Gaussian noise (AWGN). It can be observed that the noise is more visible in the dark area than in the bright area if comparing, for instance, the dark portion and the bright portion of the cloud above the bridge. This indicates that noise filtering is more necessary in the dark area than in the bright area. The lighter area can accommodate more additive noise before the noise becomes visible. This property has found application in embedding digital watermarks [huang 1998].

The direct impact that luminance masking has on image and video compression is related to quantization, which is covered in detail in Chapter 2. Roughly speaking, quantization is a process that converts a continuously distributed quantity into a set of



(a)

**FIGURE 1.8** (See color insert following page 288.)

The bridge in Vancouver: (a) Original [Courtesy of Minhuai Shi].



(b)

**FIGURE 1.8 (continued)**  
(b) Uniformly corrupted by additive white Gaussian noise (AWGN).

finitely many distinct quantities. The number of these distinct quantities (known as quantization levels) is one of the keys in quantizer design. It significantly influences the resulting bit rate and the quality of the reconstructed image and video. An effective quantizer should be able to minimize the visibility of quantization error. The contrast sensitivity function provides a guideline in analysis of the visibility of quantization error. Therefore, it can be applied to quantizer design. Luminance masking suggests a nonuniform quantization scheme that takes the contrast sensitivity function into consideration. One such example was presented in [watson 1987].

**1.2.2.2 Texture Masking**

Texture masking is sometimes also called detail dependence [connor 1972], spatial masking [netravali 1977; lim 1990], or activity masking [mitchell 1996]. It states that the discrimination threshold increases with increasing picture detail. That is, the stronger the texture, the larger the discrimination threshold. In Figure 1.8, it can be observed that the additive random noise is less pronounced in the strong texture area than in the smooth area if comparing, for instance, the dark portion of the cloud (the up-right corner of the picture) with the water area (the bottom-right corner of the picture). This is in confirmation of texture masking.

In Figure 1.9b, the number of quantization levels decreases from 256 (as in Figure 1.9a) to 16. That is, we use only 4 bits, instead of 8 bits, to represent the intensity value for each pixel. The unnatural contours, caused by coarse quantization, can be noticed in the relative uniform regions, compared with Figure 1.9a. This phenomenon was first noted in [goodall 1951] and is called false contouring [gonzalez 1992]. Now we see that the false contouring



(a)



(b)

**FIGURE 1.9** (See color insert following page 288.)  
Christmas in Winorlia: (a) Original, (b) Four-bit quantized.



(c)

**FIGURE 1.9 (continued)**

(c) Improved IGS quantized with 4 bits.

can be explained by using texture masking because texture masking indicates that the human eye is more sensitive to the smooth region than to the textured region, where intensity exhibits a high variation. A direct impact on image and video compression is that the number of quantization levels, which affects bit rate significantly, should be adapted according to the intensity variation of image regions.

### 1.2.2.3 Frequency Masking

While the above two characteristics are picture dependent in nature, frequency masking is picture independent. It states that the discrimination threshold increases with frequency increase. It is also referred to as frequency dependence.

Frequency masking can be well illustrated by using Figure 1.9. In Figure 1.9c, high-frequency random noise has been added to the original image before quantization. This method is referred to as the improved gray-scale (IGS) quantization [gonzalez 1992, p. 318]. With the same number of quantization levels (16) as in Figure 1.9b, the picture quality of Figure 1.9c is improved dramatically compared with that of Figure 1.9b: the annoying false contours have disappeared despite the increase of the root mean square value of the total noise in Figure 1.9c. This is due to the fact that the low-frequency quantization error is converted to the high-frequency noise, and that the HVS is less sensitive to the high-frequency content. We thus see, as pointed out in [connor 1972], that our human eyes function like a low-pass filter.



Owing to frequency masking, in the transform domain, say, the discrete cosine transform (DCT) domain, we can drop some high-frequency coefficients with small magnitudes to achieve data compression without noticeably affecting the perception of the HVS. This leads to a technique called transform coding, discussed in Chapter 4.

#### **1.2.2.4 Temporal Masking**

Temporal masking is another picture-independent feature of the HVS. It states that it takes a while for the HVS to adapt itself to the scene when the scene changes abruptly. During this transition the HVS is not sensitive to details. The masking takes place both before and after the abrupt change. It is called forward temporal masking if it happens after the scene change; otherwise, it is referred to backward temporal masking [mitchell 1996].

This implies that one should take temporal masking into consideration when allocating data in image and video coding.

#### **1.2.2.5 Color Masking**

Digital color image processing is gaining increasing popularity due to the wide application of color images in modern life. As mentioned earlier, we are not going to cover all aspects of the perception of the HVS. Instead, we cover only those aspects related to psychovisual redundancy, thus to image and video compression. Therefore, our discussion here on color perception is by no means exhaustive.

In physics, it is known that any visible light corresponds to an electromagnetic spectral distribution. Therefore, a color, as a sensation of visible light, is an energy with an intensity as well as a set of wavelengths associated with the electromagnetic spectrum. Obviously, intensity is an attribute of visible light. The composition of wavelengths is another attribute: chrominance. There are two elements in the chrominance attribute: hue and saturation. The hue of a color is characterized by the dominant wavelength in the composition. Saturation is a measure of the purity of a color. A pure color has a saturation of 100%, whereas a white light has a saturation of 0.

##### **1.2.2.5.1 RGB Model**

The RGB primary color system is the most well known among several color systems. This is due to the following feature of the human perception of color. The color sensitive area in the HVS consists of three different sets of cones and each set is sensitive to the light of one of the three primary colors: red, green, and blue. Consequently, any color sensed by the HVS can be considered as a particular linear combination of the three primary colors. Many research results are available, the C.I.E. (Commission Internationale de l'Eclairage) chromaticity diagram being a well-known example. These results can be easily found in many classic optics and digital image processing texts.

The RGB model is used mainly in color image acquisition and display. In color signal processing including image and video compression, however, the luminance–chrominance color system is more efficient and, hence, widely used. This has something to do with the color perception of the HVS. It is known that the HVS is more sensitive to green than to red, and is least sensitive to blue. An equal representation of red, green, and blue leads to inefficient data representation when the HVS is the ultimate viewer. Allocating data only to the information that the HVS can perceive, on the other hand, can make video coding more efficient.

Luminance is concerned with the perceived brightness, while chrominance is related to the perception of hue and saturation of color. That is, roughly speaking, the luminance–chrominance representation agrees more with the color perception of the HVS. This feature

makes the luminance–chrominance color models more suitable for color image processing. A good example is presented in [gonzalez 1992], about histogram equalization. It is well known that applying histogram equalization can bring out some details originally in dark regions. Applying histogram equalization to the RGB components separately can certainly achieve the goal. In doing so, however, the chrominance elements hue and saturation have been changed, thus leading to color distortion. With a luminance–chrominance model, histogram equalization can be applied to the luminance component only. Hence, the details in the dark regions are brought out, whereas the chrominance elements remain unchanged, hence no color distortion. With the luminance component  $Y$  serving as a black–white signal, a luminance–chrominance color model offers compatibility with black and white TV systems. This is another merit of luminance–chrominance color models.

To be discussed next are several different luminance–chrominance color models: HSI, YUV, YCbCr, and YIQ.

#### 1.2.2.5.2 Gamma-Correction

It is known that a nonlinear relationship (basically a power function) exists between electrical signal magnitude and light intensity for both cameras and CRT-based display monitors [haskell 1996]. That is, the light intensity is a linear function of the signal voltage raised to the power of  $\gamma$ . It is a common practice to correct this nonlinearity before transmission. This is referred to as gamma-correction. The gamma-corrected RGB components are denoted by  $R'$ ,  $G'$ , and  $B'$ , respectively. They are used in the discussion on various color models. For the sake of notational brevity, we simply use  $R$ ,  $G$ , and  $B$  instead of  $R'$ ,  $G'$ , and  $B'$  in the following discussion, while keeping the gamma-correction in mind.

#### 1.2.2.5.3 HSI Model

In this model,  $I$  stands for the intensity component,  $H$  for the hue component, and  $S$  for saturation component. One merit of this color system is that the intensity component is decoupled from the chromatic components. As analyzed above, this decoupling usually facilitates color image processing tasks. Another merit is that this model is closely related to the way the HVS perceives color pictures. Its main drawback is the complicated conversion between RGB and HSI models. A detailed derivation of the conversion may be found in [gonzalez 1992]. Because of this complexity, the HSI model is not used in any TV systems.

#### 1.2.2.5.4 YUV Model

In this model,  $Y$  denotes the luminance component, and  $U$  and  $V$  are the two chrominance components. The luminance  $Y$  can be determined from the RGB model via the following relation:

$$Y = 0.299R + 0.587G + 0.114B. \quad (1.4)$$

It is noted that the three weights associated with the three primary colors,  $R$ ,  $G$ , and  $B$ , are not the same. Their different magnitudes reflect different responses of the HVS to different primary colors.

Instead of being directly related to hue and saturation, the other two chrominance components,  $U$  and  $V$ , are defined as color differences as follows:

$$U = 0.492(B - Y), \quad (1.5)$$

and

$$V = 0.877(R - Y). \quad (1.6)$$

In this way, the YUV model lowers computational complexity. It has been used in PAL (phase alternating line) TV systems. Note that PAL is an analog composite color TV standard and is used in most of the European countries, some Asian countries, and Australia. By composite systems, we mean both the luminance and chrominance components of the TV signals are multiplexed within the same channel. For completeness, an expression of YUV in terms of RGB is given below.

$$\begin{pmatrix} Y \\ U \\ V \end{pmatrix} = \begin{pmatrix} 0.299 & 0.587 & 0.114 \\ -0.147 & -0.289 & 0.436 \\ 0.615 & -0.515 & -0.100 \end{pmatrix} \begin{pmatrix} R \\ G \\ B \end{pmatrix}. \quad (1.7)$$

#### 1.2.2.5.5 YIQ Model

This color space has been utilized in NTSC (National Television Systems Committee) TV systems for years. Note that NTSC is an analog composite color TV standard and is used in the North America and Japan. The  $Y$  still acts as the luminance component. The two chrominance components are the linear transformation of the  $U$  and  $V$  components defined in the YUV model. Specifically,

$$I = -0.545U + 0.839V, \quad (1.8)$$

and

$$Q = 0.839U + 0.545V. \quad (1.9)$$

Substituting the  $U$  and  $V$  components expressed in Equations 1.4 and 1.5 into Equations 1.8 and 1.9, we can express YIQ directly in terms of RGB. That is,

$$\begin{pmatrix} Y \\ I \\ Q \end{pmatrix} = \begin{pmatrix} 0.299 & 0.587 & 0.114 \\ 0.596 & -0.275 & -0.321 \\ 0.212 & -0.523 & 0.311 \end{pmatrix} \begin{pmatrix} R \\ G \\ B \end{pmatrix} \quad (1.10)$$

#### 1.2.2.5.6 YDbDr Model

The YDbDr model is used in SECAM (Sequential Couleur a Memoire) TV system. Note that SECAM is used in France, Russia, and some eastern European countries. The relationship between YDbDr and RGB appears below.

$$\begin{pmatrix} Y \\ Db \\ Dr \end{pmatrix} = \begin{pmatrix} 0.299 & 0.587 & 0.114 \\ -0.450 & -0.883 & 1.333 \\ -1.333 & 1.116 & -0.217 \end{pmatrix} \begin{pmatrix} R \\ G \\ B \end{pmatrix} \quad (1.11)$$

where

$$Db = 3.059U, \quad (1.12)$$

and

$$Dr = -2.169V. \quad (1.13)$$

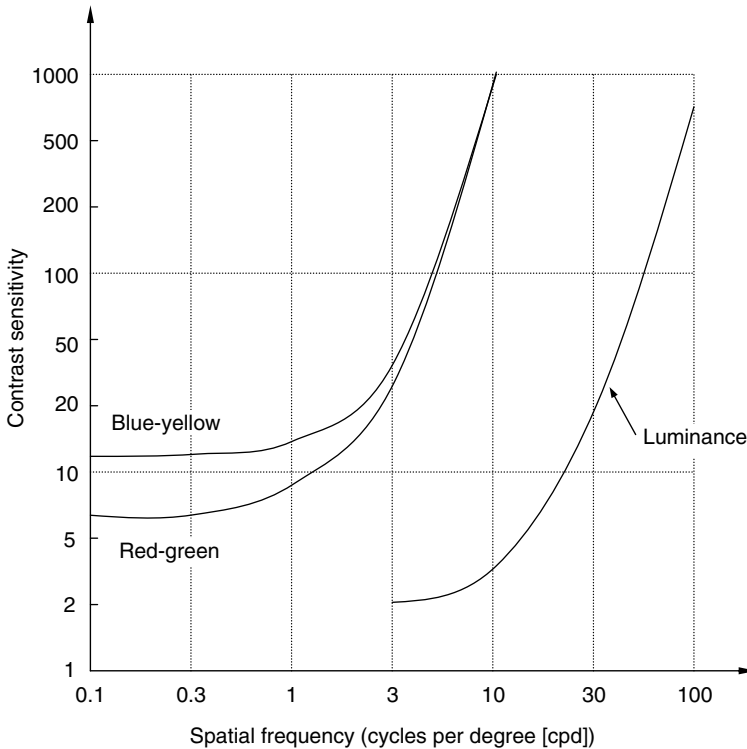
### 1.2.2.5.7 YCbCr Model

From the above, we can see that the  $U$  and  $V$  chrominance components are differences between the gamma-corrected color  $B$  and the luminance  $Y$ , and the gamma-corrected  $R$  and the luminance  $Y$ , respectively. The chrominance component pairs  $I$  and  $Q$ , and  $D_b$  and  $D_r$  are both linear transforms of  $U$  and  $V$ . Hence they are very closely related to each other. It is noted that  $U$  and  $V$  may be negative as well. To make chrominance components nonnegative, the  $Y$ ,  $U$ , and  $V$  components are scaled and shifted to produce the YCbCr model, which is used in the international coding standards JPEG and MPEG. (These two standards are covered in Chapters 7 and 16, respectively).

$$\begin{pmatrix} Y \\ Cb \\ Cr \end{pmatrix} = \begin{pmatrix} 0.257 & 0.504 & 0.098 \\ -0.148 & -0.291 & 0.439 \\ 0.439 & -0.368 & -0.071 \end{pmatrix} \begin{pmatrix} R \\ G \\ B \end{pmatrix} + \begin{pmatrix} 16 \\ 128 \\ 128 \end{pmatrix} \quad (1.14)$$

### 1.2.2.6 Color Masking and Its Application in Video Compression

It is well known that the HVS is much more sensitive to the luminance component  $Y$  than to the chrominance components  $U$  and  $V$ . Following Van Ness and Mullen [van ness 1967; mullen 1985], Mitchell, Pennebaker, Fogg, and LeGall included a figure in [mitchell 1996] to quantitatively illustrate the above statement. A modified version is shown in Figure 1.10. There, the abscissa represents spatial frequency in the unit of cycles per degree (cpd), while



**FIGURE 1.10**

Contrast sensitivity versus spatial frequency. (From Van Ness, F.I. and Bouman, M.A., *J. Opt. Soc. Am.* 57, 401, 1967; Mullen, K.T., *J. Physiol.*, 359, 381, 1985.)

the ordinate is the contrast sensitivity defined for the sinusoidal testing signal. Two observations are in order. First, for each of the three curves, i.e., curves for the luminance component  $Y$  and the chrominance components  $U$  and  $V$ , the contrast sensitivity increases when spatial frequency increases, in general. This agrees with frequency masking discussed above. Second, for the same contrast sensitivity, we see that the luminance component corresponds to a much higher spatial frequency. This is an indication that the HVS is highly sensitive to luminance than to chrominance. This statement can also be confirmed, perhaps more easily, by examining those spatial frequencies at which all three curves have data available. Then we can see that the contrast sensitivity of luminance is much lower than that of the chrominance components.

The direct impact of color masking on image and video coding is that by utilizing this psychovisual feature, we can allocate more bits to the luminance component than to the chrominance components. This leads to a common practice in color image and video coding: using full resolution for the intensity component, while using a  $2 \times 1$  subsampling both horizontally and vertically for the two chrominance components. This has been adopted in related international coding standards, discussed in Chapter 16.

#### 1.2.2.7 Summary: Differential Sensitivity

In this section, we have discussed luminance, texture, frequency, temporal, and color maskings. Before we enter Section 1.3, let us summarize what we have discussed so far.

We see that luminance masking, also known as contrast masking, is of fundamental importance among several types of masking. It states that the sensitivity of the eyes to a stimulus depends on the intensity of another stimulus. Thus it is a differential sensitivity. Both texture (detail or activity) and frequency of another stimulus significantly influence this differential sensitivity. The same mechanism exists in color perception, where the HVS is highly sensitive to luminance than to chrominance. Therefore, we conclude that differential sensitivity is the key in studying human visual perception.

These features can be utilized to eliminate psychovisual redundancy, and thus compress image and video data.

It is also noted that variable quantization, which depends on activity and luminance in different regions, seems to be reasonable from a data compression point of view. Its practical applicability, however, is somehow questionable. That is, some experimental work does not support this expectation [mitchell 1996].

It is noted that this differential sensitivity feature of the HVS is common to human perception. For instance, there is also forward and backward temporal masking in human audio perception.

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### 1.3 Visual Quality Measurement

As the definition of image and video compression indicates, image and video quality is an important factor in dealing with image and video compression. For instance, in evaluating two different compression methods, we have to base the evaluation on some definite image and video quality. When both methods achieve the same quality of reconstructed image and video, the one that requires less data is considered to be superior to the other. Alternatively, with the same amount of data, the method providing a higher quality reconstructed image or video is considered the better method. Note that here we have not considered other performance criteria, such as computational complexity.

Surprisingly, however, it turns out that the measurement of image and video quality is not straightforward. There are two types of visual quality assessment. One is objective assessment (using electrical measurements), and the other is subjective assessment (using human observers). Each has its own merits and demerits. A combination of these two methods is now widely utilized in practice. In this section, we will first discuss subjective visual quality measurement, followed by objective quality measurement.

### 1.3.1 Subjective Quality Measurement

It is natural that the visual quality of reconstructed video frames should be judged by human viewers if they are to be the ultimate receivers of the data (see Figure 1.1). Therefore, the subjective visual quality measure plays an important role in visual communications.

In subjective visual quality measurement, a set of video frames are generated with varying coding parameters. Observers are invited to subjectively evaluate the visual quality of these frames. Specifically, observers are asked to rate the pictures by giving some measure of picture quality. Alternatively, observers are requested to provide some measure of impairment to the pictures. A five-scale rating system of the degree of impairment, used by Bell Laboratories, is listed below [sakrison 1979]. It has been adopted as one of the standard scales in CCIR Recommendation 500-3 [CCIR 1986]. Note that CCIR is now ITU-R (International Telecommunications Union-Recommendations).

1. Impairment is not noticeable.
2. Impairment is just noticeable.
3. Impairment is definitely noticeable, but not objectionable.
4. Impairment is objectionable.
5. Impairment is extremely objectionable.

In the subjective evaluation, there are a few things worth mentioning. In most applications, there is a whole array of pictures simultaneously available for evaluation. These pictures are generated with different encoding parameters. By keeping some parameters fixed while making one parameter (or a subset of parameters) free to change, the resulting quality rating can be used to study the effect of the one parameter (or the subset of parameters) on encoding. An example using this method to study the effect of varying numbers of quantization levels on image quality can be found in [gonzalez 1992].

Another possible way to study subjective evaluation is to identify pictures with the same subjective quality measure from the whole array of pictures. From this subset of test pictures, we can produce, in the encoding parameter space, isopreference curves that can be used to study the effect of the parameter(s) under investigation. An example using this method to study the effect of varying both image resolution and numbers of quantization levels on image quality can be found in [huang 1965].

In the rating, a whole array of pictures is usually divided into columns with each column sharing some common conditions. The evaluation starts within each column with a pairwise comparison. This is because pairwise comparison is relatively easy for the eyes. As a result, pictures in one column are arranged in an order according to visual quality, and quality or impairment measures are then assigned to the pictures in the one column. After each column has been rated, unification between columns is necessary. That is, different columns need to have a unified quality measurement. As pointed out in [sakrison 1979], this task is not easy because it means we may need to equate impairment that results from different types of errors.

One thing is understood from the above discussion: Subjective evaluation of visual quality is costly, and it needs large number of pictures and observers. The evaluation takes a long time because human eyes are easily fatigued and bored. Some special measures have to be taken to arrive at an accurate subjective quality measure. Examples in this regard include averaging subjective ratings and taking their deviation into consideration. For further details on subjective visual quality measurement, readers may refer to [sakrison 1979; hidaka 1990; webster 1993].

### 1.3.2 Objective Quality Measurement

In this section, we first introduce the concept of signal to noise ratio (SNR), which is a popularly utilized objective quality assessment. Then we present a promising new objective visual quality assessment technique based on human visual perception.

#### 1.3.2.1 Signal to Noise Ratio

Consider Figure 1.11, where  $f(x, y)$  is the input image to a processing system. The system can be a low-pass filter, a subsampling system, or a compression system.

It can even represent a process in which AWGN corrupts the input image. The  $g(x, y)$  is the output of the system. In evaluating the quality of  $g(x, y)$ , we define an error function  $e(x, y)$  as the difference between the input and the output. That is,

$$e(x, y) = f(x, y) - g(x, y). \quad (1.15)$$

The mean square error,  $E_{ms}$ , is defined as

$$E_{ms} = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} e(x, y)^2, \quad (1.16)$$

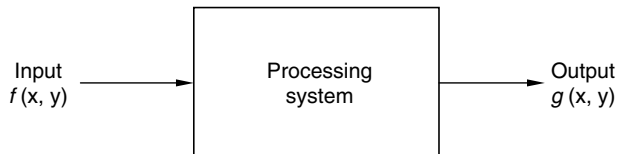
where  $M$  and  $N$  are the dimensions of the image in the horizontal and vertical directions. Note that it is sometimes denoted by MSE. The root mean square error,  $E_{rms}$ , is defined as

$$E_{rms} = \sqrt{E_{ms}}. \quad (1.17)$$

It is sometimes denoted by RMSE.

As noted earlier, SNR is widely used in objective quality measurement. Depending whether mean square error or root mean square error is used, the SNR may be called the mean square signal to noise ratio,  $SNR_{ms}$ , or the root mean square signal to noise ratio,  $SNR_{rms}$ . We have

$$SNR_{ms} = 10 \log_{10} \left( \frac{\sum_{x=0}^{M-1} \sum_{y=0}^{N-1} g(x, y)^2}{MN \cdot E_{ms}} \right), \quad (1.18)$$



**FIGURE 1.11**  
An image processing system.

and

$$SNR_{rms} = \sqrt{SNR_{ms}}. \quad (1.19)$$

In image and video data compression, another closely related term, *PSNR* (peak signal to noise ratio), which is essentially a modified version of  $SNR_{ms}$ , is widely used. It is defined as follows:

$$PSNR = 10 \log_{10} \left( \frac{255^2}{E_{ms}} \right) \quad (1.20)$$

The interpretation of the SNR is that the larger the SNR ( $SNR_{ms}$ ,  $SNR_{rms}$ , or  $PSNR$ ) the better the quality of the processed image,  $g(x, y)$ . That is, the closer the processed image  $g(x, y)$  is to the original image  $f(x, y)$ . This seems correct; however, from our earlier discussion about the features of the HVS, we know that the HVS does not respond to visual stimuli in a straightforward way. Its low-level processing unit is known to be nonlinear. Several masking phenomena exist. Each confirms that the visual perception of the HVS is not simple. It is worth noting that our understanding of the high-level processing unit of the HVS is far from complete. Therefore, we may understand that the SNR does not always provide us with reliable assessments of image quality. One good example is presented in Section 1.2.2.3, which uses the IGS quantization technique to achieve high compression (using only four bits for quantization instead of the usual eight bits) without introducing noticeable false contouring. In this case, the subjective quality is high, and the SNR decreases due to low-frequency quantization noise and additive high-frequency random noise. Another example drawn from our discussion about the masking phenomena is that some additive noise in bright areas or in highly textured regions may be masked, whereas some minor artifacts in dark and uniform regions may turn out to be quite annoying. In this case, the SNR cannot truthfully reflect visual quality.

On the one hand, we see that the objective quality measure does not always provide reliable picture quality assessment. On the other hand, however, its implementation is much faster and easier than that of the subjective quality measure. Furthermore, objective assessment is repeatable. Owing to these merits, objective quality assessment is still widely used despite this drawback.

It is noted that combining subjective and objective assessment has been a common practice in international coding-standard activity.

### 1.3.2.2 An Objective Quality Measure Based on Human Visual Perception

Introduced here is a new development in visual quality assessment, which is an objective quality measurement based on human visual perception [webster 1993]. Since it belongs to the category of objective assessment, it possesses merits, such as repeatability, and fast and easy implementation. On the other hand, based on human visual perception, its assessment of visual quality agrees closely to that of subjective assessment. In this sense, the new method attempts to combine the merits of the two different types of assessment.

#### 1.3.2.2.1 Motivation

Visual quality assessment is best conducted via the subjective approach because in this case, the HVS is the ultimate viewer. The implementation of subjective assessment is, however, time-consuming, costly, and it lacks repeatability. On the other hand, although not always accurate, objective assessment is fast, easy, and repeatable. The motivation here



is to develop an objective quality measurement system such that its quality assessment is very close to that obtained by using subjective assessment. To achieve this goal, this objective system is based on subjective assessment. That is, it uses the rating achieved via subjective assessment as a criterion to search for new objective measurements so as to have the objective rating as close to the subjective one as possible.

### 1.3.2.2.2 Methodology

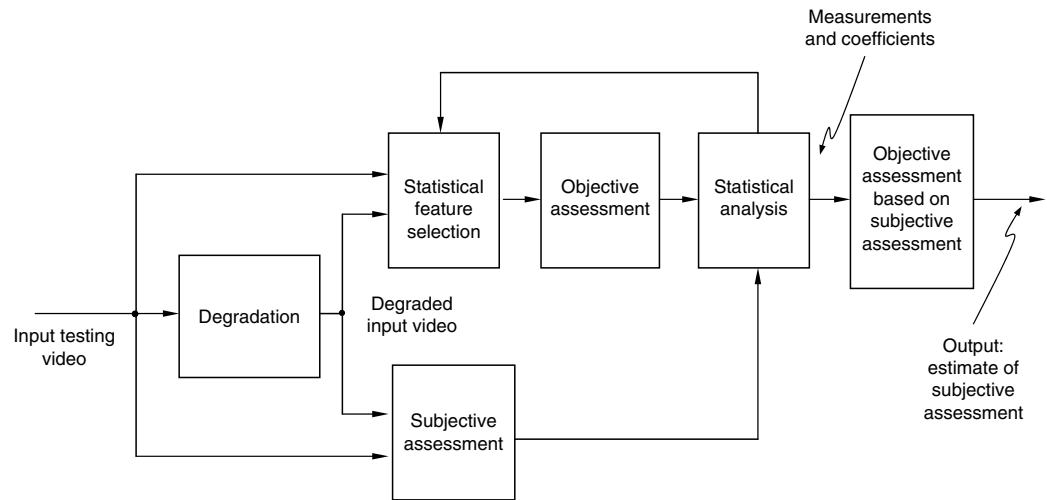
The derivation of the objective quality assessment system is shown in Figure 1.12. The input testing video goes through a degradation block, resulting in degraded input video. The degradation block, or impairment generator, includes various video compression codecs (coder–decoder pairs) with bit rates ranging from 56 kbits/s to 45 Mbits/s, and other video operations. The input video and degraded input video form a pair of testing video, which is sent to a subjective assessment block as well as a statistical feature selection block.

A normal subjective visual quality assessment, as introduced in Section 1.3.2.2.2, is performed in the subjective assessment block, which involves a large panel of observers (e.g., 48 observers in [webster 1993]). In the statistical feature selection block, a variety of statistical operations are conducted and various statistical features are selected. Examples cover Sobel filtering, Laplacian operator, first-order differencing, moment calculation, fast Fourier transform, etc. Statistical measurements are then selected based on these statistical operations and features. An objective assessment is formed as follows:

$$\hat{s} = a_0 + \sum_{i=1}^l a_i n_i, \quad (1.21)$$

where  $\hat{s}$  denotes the output rating of the object assessment, or simply the objective measure, which is supposed to be a good estimate of the corresponding subjective score. The  $n_i, i = 1, \dots, l$  are selected objective measurements. The  $a_0, a_i, i = 1, \dots, l$  are coefficients in the linear model of the objective assessment.

The results of the objective and subjective assessments are applied to a statistical analysis block. In the statistical analysis block, the objective assessment rating is compared with the subjective assessment rating. The result of the comparison is fed back to the statistical



**FIGURE 1.12**  
Block diagram of objective assessment based on subjective assessment.

feature selection block. The statistical measurements obtained in the statistical feature selection block are examined according to their performance in the assessment. A statistical measurement is regarded to be good if it can reduce by a significant amount the difference between the objective and the subjective assessments. The best measurement is determined via an exhaustive search among the various measurements. Note that the coefficients in Equation 1.21 are also examined in the statistical analysis block in a similar manner to that used for the measurements.

The measurements and coefficients determined after iterations result in an optimal objective assessment via Equation 1.21, which is finally passed to the last block as the output of the system. The whole process will become much clearer with the explanation provided below.

#### 1.3.2.2.3 Results

The results reported in [webster 1993] are introduced here.

#### 1.3.2.2.4 Information Features

As mentioned in Section 1.2.2, differential sensitivity is a key in human visual perception. Two selected features, perceived spatial information  $[SI]$  (the amount of spatial detail) and perceived temporal information  $[TI]$  (the amount of temporal luminance variation), involve pixel differencing.  $SI$  is defined as shown:

$$SI(f_n) = STD_s\{Sobel(f_n)\}, \quad (1.22)$$

where

$STD_s$  denotes standard deviation operator in the spatial domain

$Sobel$  denotes the Sobel operation

$f_n$  denotes the  $n$ th video frame

Temporal information is defined similarly:

$$TI(f_n) = STD_s\{\Delta f_n\}, \quad (1.23)$$

where  $\Delta f_n = f_n - f_{n-1}$ , i.e., the successive frame difference.

#### 1.3.2.2.5 Determined Measurements

The parameter  $l$  in Equation 1.21 is chosen as three. That is,

$$\hat{s} = a_0 + a_1 n_1 + a_2 n_2 + a_3 n_3. \quad (1.24)$$

The measurements  $n_1$ ,  $n_2$ , and  $n_3$  are formulated based on the above-defined information features:  $SI$  and  $TI$ .

Measurement  $n_1$

$$n_1 = RMS_t \left( 5.81 \left| \frac{SI(of_n) - SI(df_n)}{SI(of_n)} \right| \right), \quad (1.25)$$

where  $RMS_t$  represents the root mean square value taken over the time dimension;  $of_n$  and  $df_n$  denote the original  $n$ th frame and the degraded  $n$ th frame, respectively. It is observed that  $n_1$  is a measure of the relative change in the  $SI$  between the original frame and the degraded frame.

Measurement  $n_2$

$$n_2 = \mathfrak{S}_t \{0.108 \cdot \text{MAX} \{ [TI(\text{of}_n) - TI(\text{df}_n)], 0 \} \}, \quad (1.26)$$

where

$$\mathfrak{S}_t \{y_t\} = \text{STD}_t \{ \text{CONV}(y_t, [-1, 2, -1]) \}, \quad (1.27)$$

where  $\text{STD}_t$  denotes the standard deviation operator with respect to time, and  $\text{CONV}$  indicates the convolution operation between its two arguments. It is understood that  $TI$  measures temporal luminance variation (temporal motion) and the convolution kernel,  $[-1, 2, -1]$ , enhances the variation due to its high-pass filter nature. Therefore,  $n_2$  measures the difference of  $TI$  between the original and degraded frames.

Measurement  $n_3$ :

$$n_3 = \text{MAX}_t \left\{ 4.23 \cdot \log_{10} \left[ \frac{TI(\text{df}_n)}{TI(\text{of}_n)} \right] \right\}, \quad (1.28)$$

where  $\text{MAX}_t$  indicates taking of the maximum value over time. Therefore, measurement  $n_3$  responds to the ratio between the  $TI$  of the degraded video and that of the original video. Distortion, such as block artifacts (discussed in Chapter 11) and motion jerkiness (discussed in Chapter 10), which occurs in video coding, will cause  $n_3$  to be large.

#### 1.3.2.2.6 Objective Estimator

The least square error procedure is applied to testing video sequences with measurements  $n_i$ ,  $i = 1, 2, 3$ , determined above, to minimize the difference between the rating scores obtained from the subjective and the objective assessments, resulting in the estimated coefficients  $a_0$  and  $a_i$ ,  $i = 1, 2, 3$ . Consequently, the objective assessment of visual quality  $\hat{s}$  becomes

$$\hat{s} = 4.77 - 0.992n_1 - 0.272n_2 - 0.356n_3. \quad (1.29)$$

#### 1.3.2.2.7 Reported Experimental Results

It was reported that the correlation coefficient between the subjective and the objective assessment scores (an estimate of the subjective score) is in the range of 0.92 to 0.94. It is noted that a set of 36 testing scenes containing various amounts of  $SI$  and  $TI$  were used in the experiment. Hence, it is apparent that quite good performance was achieved.

Although there is surely room for further improvement, this work does open a new and promising way to assess visual quality by combining subjective and objective approaches. It is objective and thus fast and easy; and because it is based on the subjective measure, it is more accurate in terms of the high correlation to human perception. Theoretically, the  $SI$  and  $TI$  measures defined on differencing are very important. They reflect the most important aspect of human visual perception.

## 1.4 Information Theory Results

In the beginning of this chapter, it was noted that the term information is considered one of the fundamental concepts in image and video compression. We will now address some

information theory results. In this section, measure of information and the entropy of an information source are covered first. We then introduce some coding theorems, which play a fundamental role in studying image and video compression.

### 1.4.1 Entropy

Entropy is a very important concept in information theory and communications. So is it in image and video compression. We first define the information content of a source symbol. Then we define entropy as average information content per symbol for a discrete memoryless source.

#### 1.4.1.1 Information Measure

As mentioned at the beginning of this chapter, information is defined as knowledge, fact, and news. It can be measured quantitatively. The carriers of information are symbols. Consider a symbol with an occurrence probability  $p$ . Its information content (i.e., the amount of information contained in the symbol),  $I$ , is defined as follows:

$$I = \log_2 \frac{1}{p} \text{ bits} \quad \text{or} \quad I = -\log_2 p \text{ bits} \quad (1.30)$$

where the bit is a contraction of binary unit. In Equation 1.30, we set the base of the logarithmic function to equal 2. It is noted that these results can be easily converted as follows for the case where the  $r$ -ary digits are used for encoding. Hence, from now on, we restrict our discussion to binary encoding.

$$I = -\log_r 2 \cdot \log_2 p \text{ bits.} \quad (1.31)$$

According to Equation 1.30, the information contained within a symbol is a logarithmic function of its occurrence probability. The smaller the probability, the more information the symbol contains. This agrees with common sense. The occurrence probability is somewhat related to the uncertainty of the symbol. A small occurrence probability means large uncertainty. In this way, we see that the information content of a symbol is about the uncertainty of the symbol. It is noted that the information measure defined here is valid for both equally probable symbols and nonequally probable symbols [lathi 1998].

#### 1.4.1.2 Average Information per Symbol

Now consider a discrete memoryless information source. By discreteness, we mean the source is a countable set of symbols. By memoryless, we mean the occurrence of a symbol in the set is independent of that of its preceding symbol. Take a look at a source of this type that contains  $m$  possible symbols  $\{s_i, i=1, 2, \dots, m\}$ . The corresponding occurrence probabilities are denoted by  $\{p_i, i=1, 2, \dots, m\}$ . According to the discussion in Section 1.4.1.1, the information content of a symbol  $s_i$ ,  $I_i$ , is equal to  $I_i = -\log_2 p_i$  bits. Entropy is defined as the average information content per symbol of the source. Obviously, the entropy,  $H$ , can be expressed as follows:

$$H = -\sum_{i=1}^m p_i \log_2 p_i \text{ bits} \quad (1.32)$$

From this definition, we see that the entropy of an information source is a function of occurrence probabilities. It is straightforward to show that the entropy reaches the maximum when all symbols in the set are equally probable.

### 1.4.2 Shannon's Noiseless Source Coding Theorem

Consider a discrete, memoryless, stationary information source. In what is called source encoding, a code word is assigned to each symbol in the source. The number of bits in the code word is referred to as the length of the code word. The average length of code words is referred to as bit rate, expressed in the unit of bits per symbol.

Shannon's noiseless source coding theorem states that for a discrete, memoryless, stationary information source, the minimum bit rate required to encode a symbol on average is equal to the entropy of the source. This theorem provides us with a lower bound in source coding. Shannon showed that the lower bound can be achieved when the encoding delay extends to infinity. By encoding delay, we mean the encoder waits and then encodes a certain number of symbols at once. Fortunately, with finite encoding delay, we can already achieve an average code word length fairly close to the entropy. That is, we do not have to actually sacrifice bit rate much to avoid long encoding delay, which involves high computational complexity and a large amount of memory space.

Note that the discreteness assumption is not necessary. We assume a discrete source simply because digital image and video are the focus here. Stationarity assumption is necessary in deriving the noiseless source coding theorem. This assumption may not be satisfied in practice. Hence, Shannon's theorem is a theoretical guideline only. There is no doubt, however, that it is a fundamental theoretical result in information theory.

In summary, the noiseless source coding theorem, Shannon's first theorem published in his celebrated paper [Shannon 1948], is concerned with the case where both the channel and the coding system are noise free. The aim under these circumstances is coding compactness. The more compact, the better the coding. This theorem specifies the lower bound, which is the source entropy, and how to reach this lower bound.

One way to evaluate the efficiency of a coding scheme is to determine its efficiency with respect to the lower bound, i.e., entropy. The efficiency  $\eta$  is defined as follows:

$$\eta = \frac{H}{L_{\text{avg}}}, \quad (1.33)$$

where  $H$  is entropy, and  $L_{\text{avg}}$  denotes the average length of the code words in the code. As the entropy is the lower bound, the efficiency never exceeds the unity, i.e.,  $\eta \leq 1$ . The same definition can be generalized to calculate the relative efficiency between two codes. That is,

$$\eta = \frac{L_{\text{avg},1}}{L_{\text{avg},2}}, \quad (1.34)$$

where  $L_{\text{avg},1}$  and  $L_{\text{avg},2}$  represent the average code word length for code 1 and code 2, respectively. We usually put the larger of the two in the denominator, and  $\eta$  is called the efficiency of code 2 with respect to code 1. A complementary parameter of coding efficiency is coding redundancy,  $\zeta$ , which is defined as

$$\zeta = 1 - \eta. \quad (1.35)$$

### 1.4.3 Shannon's Noisy Channel Coding Theorem

If a code has an efficiency of  $\eta = 1$  (i.e., it reaches the lower bound of source encoding) then coding redundancy is  $\zeta = 0$ . Now consider a noisy transmission channel. In transmitting the coded symbol through the noisy channel, the received symbols may be erroneous due to the lack of redundancy. On the other hand, it is well known that by adding redundancy (e.g., parity check bits) some errors occurring during the transmission over the noisy channel may be identified. The coded symbols are then resent. In this way, we see that adding redundancy may combat noise.

Shannon's noisy channel coding theorem states that it is possible to transmit symbols over a noisy channel without error if the bit rate is below a channel capacity,  $C$ . That is,

$$R < C \quad (1.36)$$

where  $R$  denotes the bit rate. The channel capacity is determined by the noise and signal power.

In conclusion, the noisy channel coding theorem, Shannon's second theorem [shannon 1948], is concerned with a noisy, memoryless channel. By memoryless, we mean the channel output corresponding to the current input is independent of the output corresponding to previous input symbols. Under these circumstances, the aim is reliable communication. To be error-free, the bit rate cannot exceed channel capacity. That is, channel capacity sets an upper bound on the bit rate.

### 1.4.4 Shannon's Source Coding Theorem

As seen in Sections 1.4.2 and 1.4.3, the noiseless source coding theorem defines the lowest possible bit rate for noiseless source coding and noiseless channel transmission; whereas the noisy channel coding theorem defines the highest possible coding bit rate for error-free transmission. Therefore, both theorems work for reliable (no error) transmission. In this section, we continue to deal with discrete memoryless information sources, but we discuss the situation in which lossy coding is encountered. As a result, distortion of the information sources takes place. For instance, quantization, discussed in Chapter 2, causes information loss. Therefore, it is concluded that if an encoding procedure involves quantization, then it is lossy coding. That is, errors occur during the coding process, even though the channel is error-free. We want to find the lower bound of the bit rate for this case.

The source coding theorem [shannon 1948] states that for a given distortion  $D$ , there exists a rate distortion function  $R(D)$  [berger 1971], which is the minimum bit rate required to transmit the source with distortion less than or equal to  $D$ . That is, to have distortion not larger than  $D$ , the bit rate,  $R$ , must satisfy the following condition:

$$R \geq R(D). \quad (1.37)$$

A more detailed discussion about this theorem and the rate distortion function is given in Chapter 15, which deals with video coding.

### 1.4.5 Information Transmission Theorem

It is clear that by combining the noisy channel coding theorem and the source coding theorem, we can derive the following relationship:

$$C \geq R(D) \quad (1.38)$$

This is called the information transmission theorem [slepian 1973]. It states that if the channel capacity of a noisy channel,  $C$ , is larger than the rate distortion function  $R(D)$  then it is possible to transmit an information source with distortion  $D$  over a noisy channel.

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## 1.5 Summary

In this chapter, we first discussed the necessity for image and video compression. It is shown that image and video compression becomes an enabling technique in today's exploding number of digital multimedia applications. Then, we show that the feasibility of image and video compression rests in redundancy removal. Three types of redundancy are studied: statistical redundancy, coding redundancy, and psychovisual redundancy. Statistical redundancy comes from interpixel correlation. By interpixel correlation, we mean correlation between pixels either located in one frame (spatial or intraframe redundancy) or pixels located in successive frames (temporal or interframe redundancy). Psychovisual redundancy is based on the features (several types of masking phenomena) of human visual perception. That is, visual information is not perceived equally from human visual point of view. In this sense, some information is psychovisually redundant. Coding redundancy is related to coding technique.

The visual quality of reconstructed image and video is a crucial criterion in the evaluation of the performance of visual transmission or storage systems. Both subjective and objective assessments are discussed. A new and promising objective technique based on subjective assessment is introduced. Because it combines the merits of both types of visual quality assessment, it achieves a quite satisfactory performance. The selected statistical features reveal some possible mechanism of the human visual perception. Further study in this regard would be fruitful.

In the last section, we introduced some fundamental information theory results, relevant to image and video compression. The results introduced include information measurement, entropy, and several theorems. All the theorems assume discrete, memoryless, and stationary information sources. The noiseless source coding theorem points out that the entropy of an information source is the lower bound of coding bit rate that a source encoder can achieve. The source coding theorem deals with lossy coding applied in a noise-free channel. It states that for a given distortion,  $D$ , there is a rate distortion function,  $R(D)$ . Whenever the bit rate in the source coding is greater than  $R(D)$ , the reconstructed source at the receiving end satisfies the fidelity requirement defined by the  $D$ . The noisy channel coding theorem states that, to achieve error-free performance, the source coding bit rate must be smaller than the channel capacity. Channel capacity is a function of noise and signal power. The information transmission theorem combines the noisy channel coding theorem and the source coding theorem. It states that it is possible to have a reconstructed waveform at the receiving end, satisfying the fidelity requirement corresponding to distortion,  $D$ , if the channel capacity,  $C$ , is larger than the rate distortion function,  $R(D)$ . Although some of the assumptions on which these theorems were developed, may not be valid in complicated practical situations, these theorems provide important theoretical limits for image and video coding. They can also be used for evaluation of the performance of different coding techniques.

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## Exercises

1. Using your own words, define spatial and temporal redundancy, and psychovisual redundancy, and state the impact they have on image and video compression.

2. Why is differential sensitivity considered the most important feature in human visual perception?
3. From the description of the newly developed objective assessment technique based on subjective assessment (Section 1.3), what points do you think are related to and support the statement made in problem 2?
4. Using your own words, interpret Weber's law.
5. What is the advantage possessed by color models that decouple the luminance component from chrominance components.
6. Why has the HIS model not been adopted by any TV systems?
7. What is the problem with the objective visual quality measure of PSNR?

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